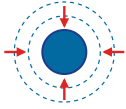


**The Big Idea** S8P5

Gravity, electricity, and magnetism are 3 major forces that act in nature. All three can work between objects **WITHOUT TOUCHING** — they work through invisible **FIELDS**!

**Fields: Forces Without Contact** S8P5a

A **field** is the invisible region around an object where it can exert force on other objects — even if they aren't touching! Think of it like an invisible "bubble of influence."

**GRAVITATIONAL FIELD**

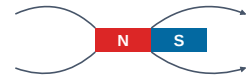
Pulls objects with mass **TOWARD** each other. Always **ATTRACTIVE**.

Example: Earth's gravity keeps you on the ground and the Moon in orbit.

**ELECTRIC FIELD**

Force between charges. Can **ATTRACT** (+/-) or **REPEL** (+/+ or -/-).

Example: Static shock; balloon sticks to a wall.

**MAGNETIC FIELD**

Force between magnets or magnetic materials. Can **ATTRACT** or **REPEL**.

Example: Magnets on a fridge; compass pointing north.



Evidence that fields exist: You can **SEE** their effects without contact — iron filings line up near a magnet, a feather falls due to Earth's gravity, static makes your hair stand up.

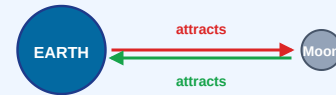
**Gravitational Force Between Objects** S8P5a

Gravity is the force of attraction between any two objects that have mass. It's **ALWAYS** attractive — never repulsive!

Two Factors Affect Gravity:

- **MASS:** More mass = **STRONGER** gravity. That's why you feel Earth's pull but not your pencil's!
- **DISTANCE:** Greater distance = **WEAKER** gravity. Gravity drops off fast as objects move apart.

★ **Examples:** Sun's huge mass keeps planets orbiting. The Moon (lots of mass, somewhat close) causes Earth's tides. You're pulled to Earth's center.

**Static Electricity & Electric Charges** S8P5b

All matter is made of atoms with **protons (+)**, **neutrons (no charge)**, and **electrons (-)**. Objects become charged when they **GAIN** or **LOSE** electrons (protons don't move).

- Object **GAINS electrons** → becomes **NEGATIVELY** charged
- Object **LOSES electrons** → becomes **POSITIVELY** charged
- **Like charges REPEL** (+/+ or -/-); **Opposite charges ATTRACT** (+/-)

Conductors vs. Insulators

CONDUCTORS — charges move **EASILY** through these.

Examples: copper wire, most metals, saltwater, graphite

INSULATORS — charges can't easily move through these.

Examples: rubber, plastic, glass, wood, dry air

**3 Ways to Charge an Object** S8P5b**1. FRICTION (rubbing)**

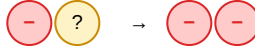
Rub two objects together; electrons **TRANSFER** from one to the other.



Example: Rub a balloon on hair → balloon gets electrons (-); hair becomes + → they attract each other!

2. CONDUCTION (touching)

A charged object **TOUCHES** another — electrons transfer directly. Both end up with the **SAME** charge.



Example: Touch a doorknob with a charged hand → both become charged the same way.

3. INDUCTION (no touch)

A charged object comes **CLOSE** (but doesn't touch). Its field pushes charges in the other object to **REARRANGE**.



Example: A charged balloon held near a wall pushes electrons in the wall away — that's why it sticks!



STATIC DISCHARGE: When built-up charge suddenly **JUMPS** to another object (through air!). Small discharge = a shock when you touch a doorknob. Big discharge = **lightning!**

ACTIVITY 1: Fields, Gravity & Charges

1. Which of Earth's properties would **DOUBLE** its gravitational pull on you — doubling its mass or moving you twice as far away? _____
2. If an object **LOSES** electrons, its charge becomes _____ (positive / negative).
3. Two + charges near each other will _____ (attract / repel).
4. Rubbing a balloon on your head charges it by _____ (friction / conduction / induction).
5. Which is a conductor: wood or copper? _____

⚡ Flip over for magnets, electromagnets, motors, generators & Faraday's Law →

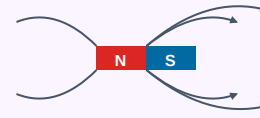
Magnetism Basics S8P5a

Rules of Magnets

- Every magnet has **TWO POLES**: North (N) and South (S)
- Like poles REPEL** (N–N or S–S push apart) 🚫
- Opposite poles ATTRACT** (N–S pull together) 🤝
- Poles **CAN'T** be separated — cut a magnet in half, you get 2 smaller magnets each with N & S ends
- Magnetic materials**: iron, nickel, cobalt
- Earth is a giant magnet!** That's why a compass works.

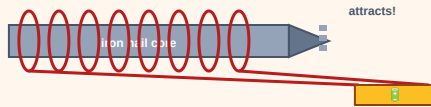
Distance & Field Strength

Magnetic force is **STRONGEST** near the poles and **WEAKER** as you move away. Field lines run from **N → S** outside the magnet.



Electromagnets S8P5c

An **electromagnet** is a temporary magnet made by running electric current through a coil of wire. It's **ONLY** magnetic when the current flows!



Making Electromagnets Stronger

Four ways to increase the strength:

- MORE wire TURNS (coils)** around the core
- BIGGER battery / more dry cells** (more current)
- LARGER iron core**
- SHORTER distance** between magnet and object

⚡ **Real-world uses**: junkyard cranes, MRI machines, doorbells, speakers, electric bells. **Advantage**: you can turn them ON and OFF!

Electric Motors & Generators S8P5c

Both use the link between **electricity** and **magnetism** — but they work in **OPPOSITE** directions!

ELECTRIC MOTOR

ELECTRICITY → MOTION

Current flows through a coil inside a magnetic field → coil **SPINS**.



Uses: fans, blenders, electric cars, power tools

ELECTRIC GENERATOR

MOTION → ELECTRICITY

Spinning a coil in a magnetic field **CREATES** current (electromagnetic induction).



Uses: power plants (wind, hydro, coal, nuclear), bike lights, emergency generators

Faraday's Law (Electromagnetic Induction)

Michael Faraday discovered that a **CHANGING** magnetic field near a wire causes electric current to flow in that wire. This is how generators work!

🔑 **Key idea**: Move a magnet through a coil → the changing magnetic field creates (INDUCES) current in the wire. Move it **FASTER** = **MORE** current!

Faraday's Law factors that INCREASE induced current:

- Moving the magnet **FASTER**
- Using a **STRONGER** magnet
- Using **MORE** coils of wire



ACTIVITY 2: Electromagnets & Motors

- Name **TWO** ways to make an electromagnet stronger: _____ and _____
- A fan is a _____ (motor/generator) — converts _____ → _____ energy.
- A wind turbine is a _____ (motor/generator) — converts _____ → _____ energy.
- If you move a magnet through a coil **FASTER**, how does the induced current change? _____

ACTIVITY 3: Fields Comparison

Label each scenario (gravity / electric / magnetic):

- Compass needle points north: _____
- An apple falls from a tree: _____
- Hair sticks up after rubbing a balloon on it: _____
- The Moon orbits Earth: _____
- Fridge magnet stays stuck: _____
- Static shock from a doorknob: _____

I Can... (Self-Check for Milestones)

- | | |
|--|--|
| <ul style="list-style-type: none"> ☐ Describe how fields allow forces without contact (S8P5a) ☐ Identify gravitational, electric, and magnetic fields ☐ Explain how mass and distance affect gravity ☐ Identify + and – charges and what attracts/repels ☐ Distinguish conductors from insulators (S8P5b) ☐ Describe charging by friction, conduction, and induction | <ul style="list-style-type: none"> ☐ Explain static discharge (including lightning) ☐ Describe magnetic poles and field lines ☐ Explain how an electromagnet works ☐ List factors that affect electromagnet strength (S8P5c) ☐ Distinguish electric motors (E → M) from generators (M → E) ☐ Apply Faraday's Law (moving magnet induces current) |
|--|--|